

Upper Extremity COAR

Objectives UE COAR

Clinical Skills

1. Identify important aspects of a HPI relevant to a patient who presents with shoulder pain
2. Integrate osteopathic diagnosis into a focused physical exam relevant for a specific complaint, in this case, shoulder pain
3. Create an osteopathic treatment plan for the patient with shoulder and/or upper extremity pain
4. Describe red flags related to shoulder or upper extremity pain

Osteopathic Manipulative Medicine

1. Discuss the A (Autonomics), B (Biomechanics), C (Circulation), and S (Screening) aspects of how structural findings could contribute to Shoulder and/or upper extremity pain
2. Prescribe OMT to improve the anatomic and physiologic functioning of the patient as part of a treatment plan
3. Perform osteopathic manipulative treatment aimed at assisting a patient who presents with shoulder or upper extremity pain
4. Identify indications or contraindications to the use of OMT in shoulder or upper extremity

Anatomy

1. Review dermatomes and muscular structures of the shoulder and arm
2. Consider the importance of multiple joints of the shoulder including the glenohumeral joint, the acromioclavicular joint, the sternoclavicular joint, and the scapulothoracic joint

Radiology

1. Review relevant shoulder anatomic structures on X-ray and MRI
2. Define imaging criteria for shoulder pain

Pharmacology

1. Understand pharmacological interventions for shoulder complaint(s)

History

Location: Clinic setting

CC: 42 yo Caucasian female with PMHX of HTN presents for right anterior shoulder pain and radiating arm pain of 4 months duration

HPI:

- **Location:** Right anterior shoulder- sometimes can feel it on the top of the shoulder (trapezius region) and into the right side of the neck
- **Onset-** started about 4 months ago- insidious onset, no trauma accident or injury- she did start playing tennis more- progressively getting worse- can keep her awake at night- and she does not want to lie on her right side- it really hurts.
- **Provocation:** reaching overhead hurts sometimes- and she's cut back on playing tennis because that can make it really sore- it kind of has a low-level ache all the time
- **Palliation:** not using it makes it feel a little better- she's been using Ibuprofen which does help- 800 mg three times daily (TID) to four times daily (QID)
- **Quality:** 5/10 at worst, 1-2/10 at least, currently 3/10 sore and aching, can be sharp when she lifts her arm up.
- **Radiation:** "It feels like it can radiate down the front of my arm sometimes- to the outer aspect of my elbow and into my forearm, as well as the right lower neck"
- **Associated symptoms:** sometimes her right shoulder feels tired or weak, and she has a strange new pain in upper back mid back (points between scapula)- which is really really sore, and started about 2 days ago
- **Timing:** No better or worse at any time of day
- **Previous occurrence:** I've never had this pain before- She's had no prior evaluation- no prior imaging or PT

PMHX: HTN, migraine headaches, GERD

Allergies: penicillin- hives, Vicodin- nausea, No allergies to latex or food

PSHX: Cholecystectomy, Tonsillectomy, Right Carpal Tunnel release 3 years ago

Trauma history: fractured right clavicle and AC Joint Separation car accident- non-surgical- 5 years ago

Medications: Lisinopril 10 mg QD, Ibuprofen 800mg TID- QID most days over the last few months

Social History: smokes ¼ ppd, denies recreational drug use, 1 glass of wine weekly, married, 2 children, CNA, walks and plays tennis for exercise, 1 cup of coffee per day

Family History: M: alive HTN, DM; F: alive lung CA, HTN; no siblings; children healthy; denies family hx of inflammatory arthritis

Patient Concerns: "Bothering me all the time now- even at work- I just want it to go away"

ROS:

General: denies fevers chills night sweats weight loss or weight gain, + trouble sleeping

MSK: admits to some mild weakness, denies other joint pain, swelling, redness, denies other myalgias or weakness, denies atrophy or fasciculations, admits to some right shoulder ROM restriction in flexion/abduction, no decrease in neck ROM or elbow ROM, denies clicking, popping, catching, or locking of shoulder or elbow or neck, admits to pain as per HPI. Admits to new onset thoracic back pain- like a dull ache.

Neuro: denies numbness or tingling, admits to some mild weakness of flexion/abduction, + radiation of pain, denies headaches or dizziness, denies left sided symptoms

GI: +GERD is getting worse, lots of heartburn and some associated nausea, denies vomiting, denies diarrhea or constipation, +mild epigastric pain

Physical Exam

Vitals: HR: 80, RR 14, BP 150/90, BMI: 28 Temp: 97.9

General:

- NAD, A&OX3, overweight female with normal gait, slouching posture, anterior head position (EAM anterior), right anterior shoulder with increased internal rotation

Neuro:

- Strength +5/5 in all myotome's bilaterally, Sensation intact in C5-T1 dermatomes bilaterally, Reflexes +2/4 equal and symmetric at triceps, biceps and brachioradialis bilaterally, Spurling's is negative bilaterally

MSK:

SHOULDER:

- Right shoulder position anterior and internally rotated, no erythema, warmth or swelling
- Tender points at Right **supraspinatus, acromioclavicular joint, supraspinatus, trapezius, levator scapulae**, Negative for tender points at right long head of biceps, short head of biceps, clavicle, sternoclavicular joint, pectoralis major, minor, deltoid, infraspinatus, rhomboids, serratus posterior superior, serratus anterior, subscapularis, teres major, minor, and bilateral cervical paraspinals.
- Right sided full active ROM with pain in full flexion above 100 degrees and aBduction at 100 degrees, slight decrease in internal rotation, symmetric motion in extension, adduction, external rotation. Painful arc above 100 degrees with scapular elevation and thoracic sidebending to accomplish motion
- Full PROM bilaterally with mild right sided tenderness in full flexion and full abduction
- Weakness on the right due to pain in full aBduction, otherwise +5/5 strength in shoulder flexion, extension, adduction, IR and EX bilaterally
- Special Tests:
 - Negative bilaterally: drop arm, Yergason's, Speeds, Apprehension, O'Brien's
 - Positive Right: pain with empty can, Apley's scratch test with IR, Neer/Hawkins

ELBOW:

- bilaterally palpation for TART negative at the radial head, olecranon, medial/lateral epicondyle, extensor/flexor wad palpation, pronator/supinator
- bilaterally AROM/PROM present, Strength 5/5 -supination/pronation, flexion and extension equal

ABDOMEN:

- normoactive BS X4, soft, mild tenderness in the epigastric region, no organomegaly, no bruit

Osteopathic exam:

- Upper crossed syndrome findings
 - Elevation of right scapula and earlier motion of scapulothoracic joint with shoulder abduction
 - Posture of shoulder protraction and IR on the right
 - Right pectoralis major hypertonicity
 - Right latissimus dorsi hypertonicity
 - Weakness of scapular stabilizers bilaterally
- Joint examination (glides and slides)
 - Scapulothoracic exam
 - Glenohumeral joint- A/P glide, Superior/inferior glide equal bilaterally
 - Right AC joint inferior
 - Right SC joint posterior and inferior
 - Right Elbow abducted (increased lateral glide)
 - Right radial Head anterior (anterior = supinated, posterior = pronated)
 - Distal radioulnar assessment with dorsal glide equal bilaterally
- Thoracic inlet- Sidebent R Rotated L
- Diaphragm- Sidebent L Rotated L
- Thoracic spine- T1FRSL, T2-3NSrRL, T4ERSL, hypertonic paraspinals, upper trapezius and levator scapulae muscles
- Cervical spine- C3ERSr, C5ERSL, C/T junction rotated left

Assessment

Differential Diagnosis (DDx): Is long! Let's consider our top 3

- Supraspinatus tendonitis
- Supraspinatus tendinosis
- Subacromial impingement
- Acromioclavicular osteoarthritis
- Glenohumeral osteoarthritis
- Subacromial bursitis
- Bicipital tendonitis
- Bicipital rupture
- Rotator cuff tear
- Labral tear
- Adhesive capsulitis (Frozen shoulder)
- Superior Labrum Anterior to Posterior (SLAP) lesion
- Somatic Dysfunction
- Cervical radiculopathy

- Visceral referral pain
 - MI
 - Gallbladder
 - Apical Lung
 - Diaphragmatic pain
- Injury/trauma/fracture
- Anterior/posterior dislocation
- Multidirectional shoulder instability
- Biceps tendinopathy/rupture
- Scapulothoracic bursitis
- Scapular instability
- Thoracic outlet syndrome
- Infection
- Rheumatologic Issue
 - Polymyalgia Rheumatica

Our top 3 Diagnoses

Adhesive capsulitis (frozen shoulder)

- History of diabetes, possible thyroid disease (also related to frozen shoulder), painful, stiff shoulder, pain is worse at night, restricted activities of daily living
- Reduced ROM in active and passive ROM in two or more planes
- Age is between 40-70 years old
- Sometimes shoulder injury precedes frozen shoulder

Subacromial Bursitis, rotator cuff tendonitis/osis or impingement

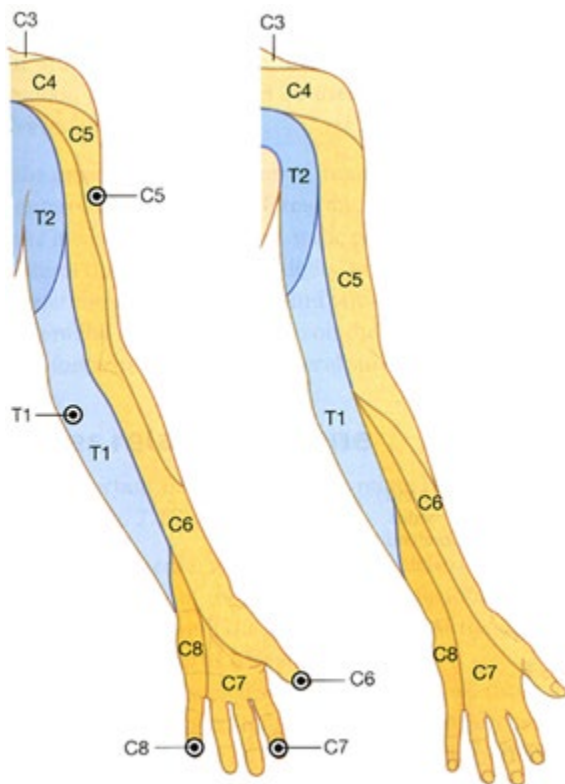
- Location of the pain (anterior), pain when sleeping on it, worse with overhead activities, limited range of active motion
- Positive Neer/Hawkins and empty can

Rotator cuff tear

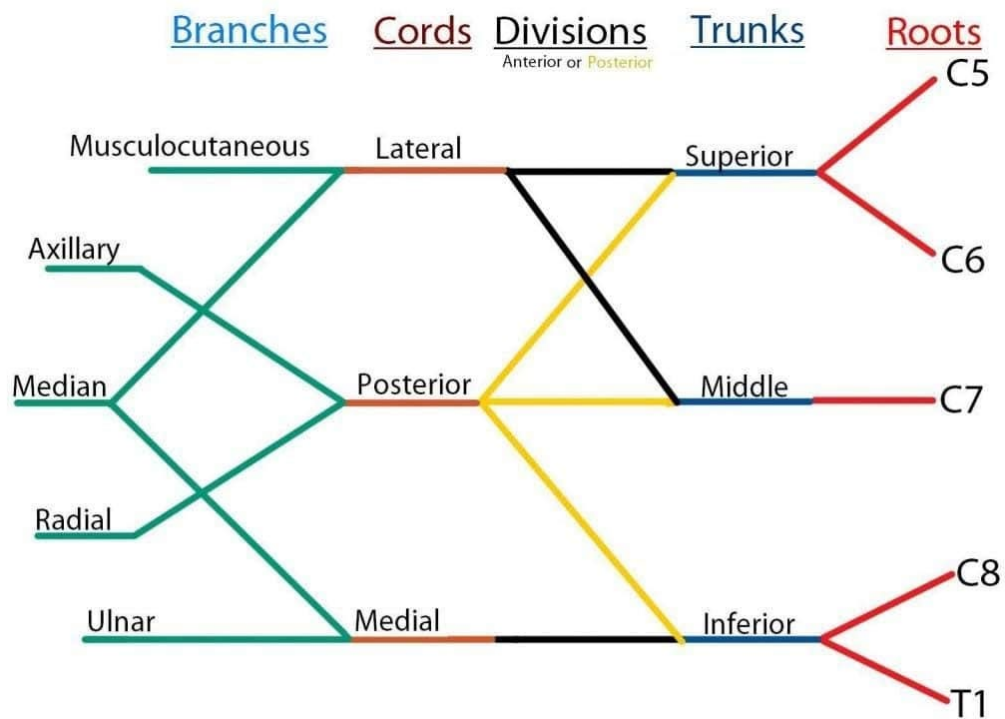
- Location of the pain (anterior), pain when sleeping on it, worse with overhead activities, limited range of active motion
- Positive Neer and empty can

Anatomy

UE Dermatomes



Brachial Plexus



Rotator Cuff Muscles/Intrinsic Muscles

Deltoid - Axillary nerve C5,6

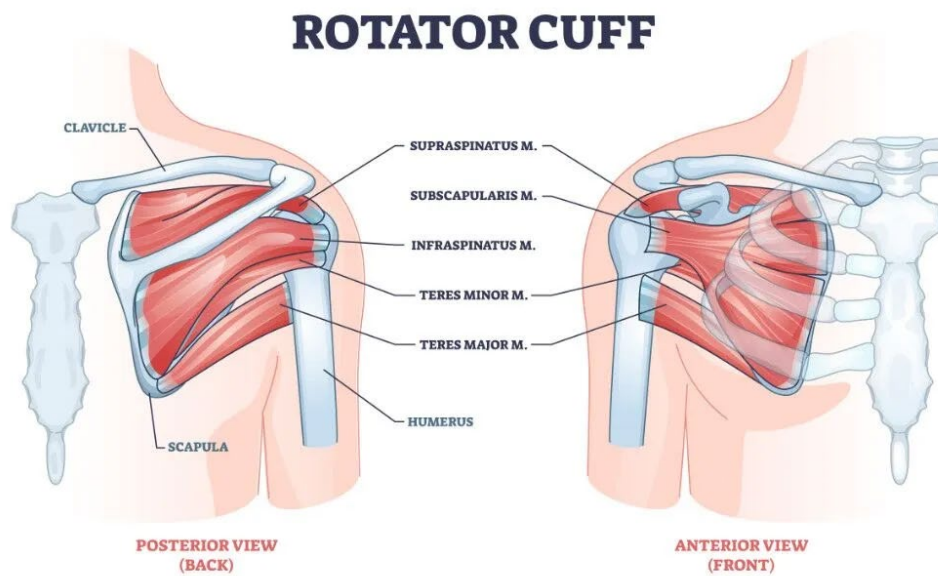
Teres major - Lower subscapular nerve C5,6

Supraspinatus - Suprascapular nerve C5,6

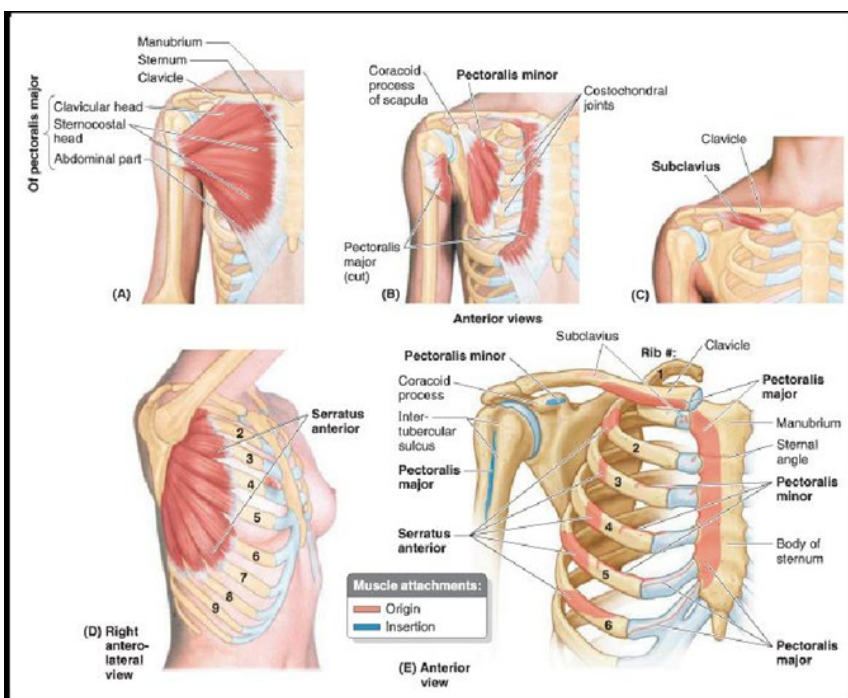
Infraspinatus - Suprascapular nerve C5,6

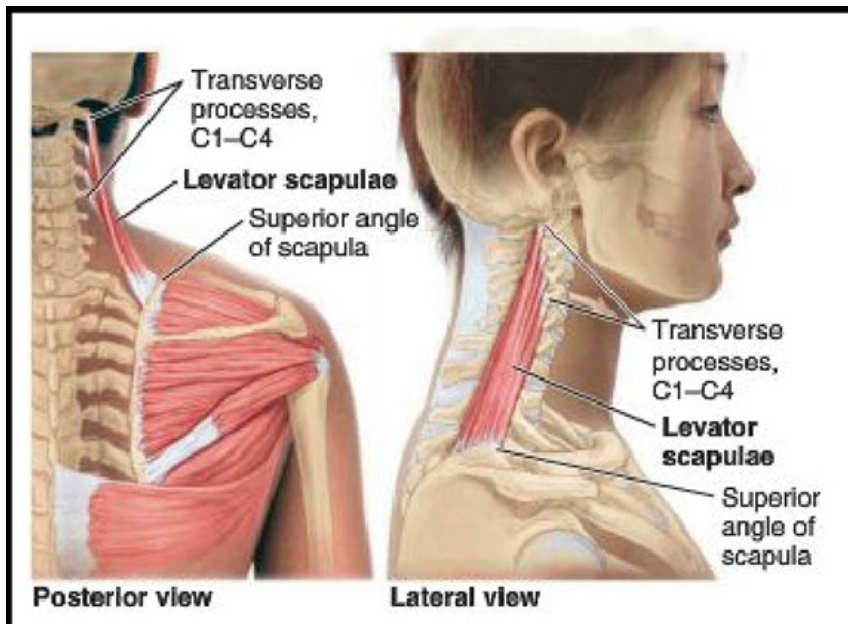
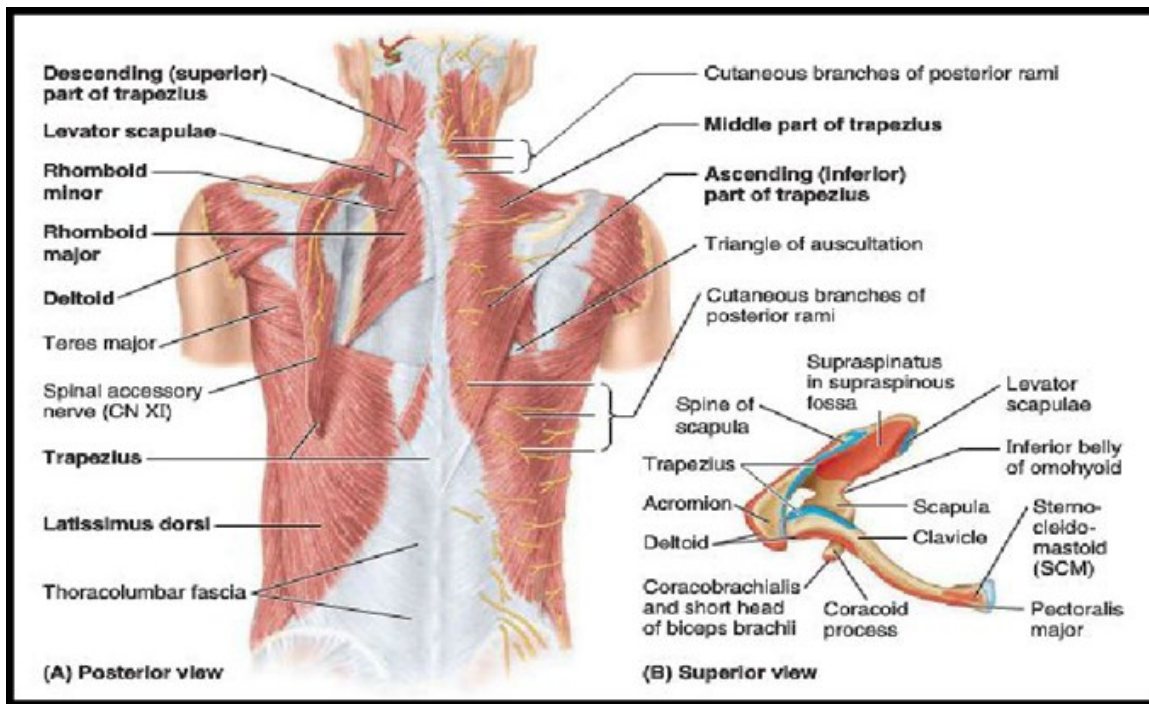
Teres minor - Suprascapular nerve C5,6

Subscapularis - Upper and lower subscapular nerve C5-7



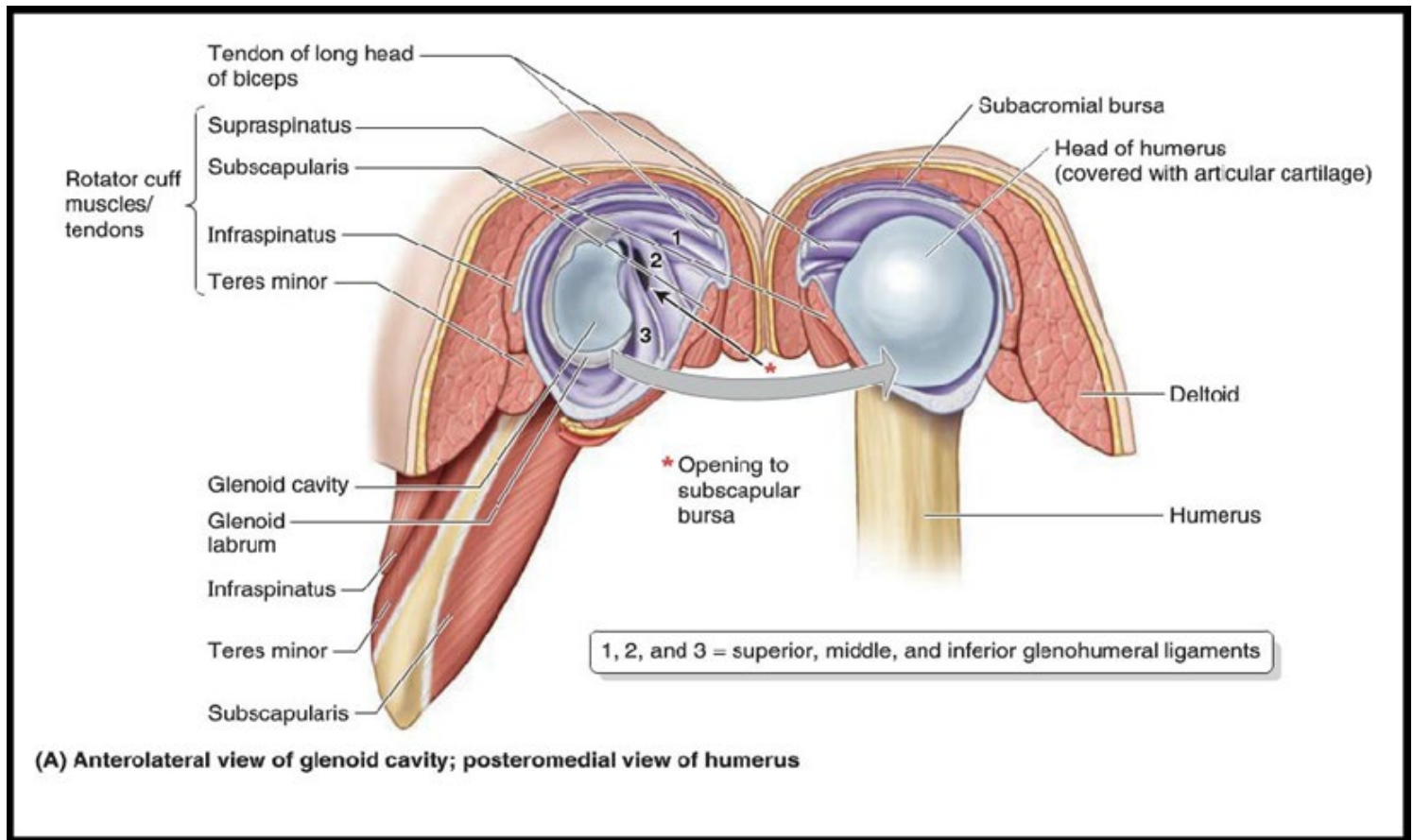
Extrinsic shoulder musculature





- **Pectoralis major** - Medial (C7-T1) and lateral (C5-6) pectoral nerves
- **Pectoralis minor** - Medial pectoral nerve (C7-T1)
- **Serratus anterior** - Long thoracic nerve (C5-7)
- **Sternocleidomastoid** - CN XI and C2-4
- **Trapezius** - Accessory nerve CN XI and ventral rami of C2-4
- **Latissimus Dorsi** - Thoracodorsal nerve C6-7
- **Levator scapulae** - Dorsal scapular nerve C5 and ventral rami C2-4
- **Rhomboids** - Dorsal scapular nerve C5

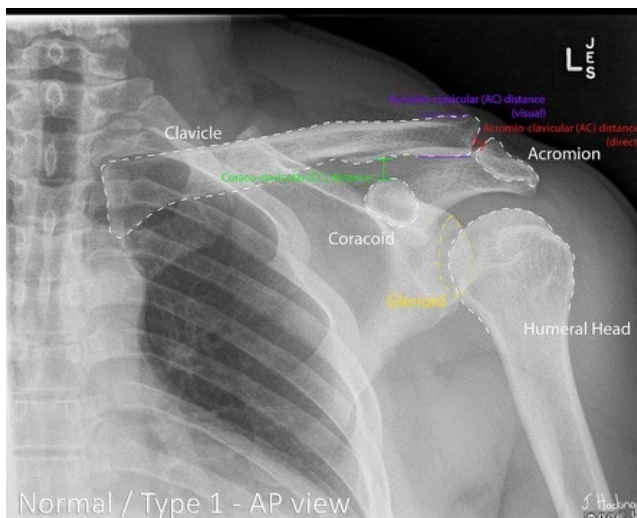
Glenohumeral Joint

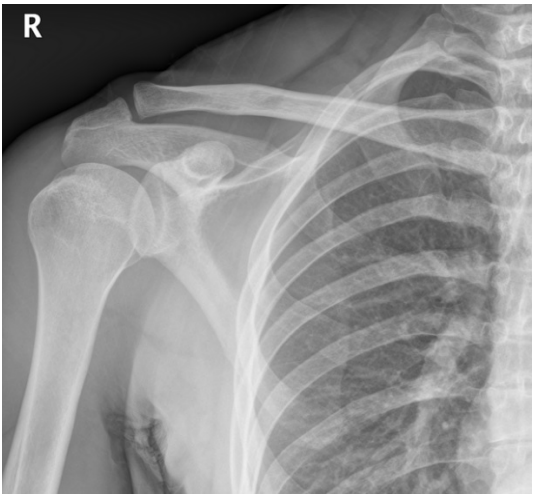


Radiology

Adhesive Capsulitis (Frozen Shoulder)

per UpToDate: "Plain radiographs are of limited diagnostic use in patients with frozen shoulder; most often the plain film is normal, except for osteopenia in some cases. Nevertheless, it is reasonable to obtain routine shoulder radiographs (including posteroanterior, external rotation, Y-outlet, and axillary views) in patients with suspected frozen shoulder to exclude other disorders, such as glenohumeral osteoarthritis, which may present with similar symptoms and clinical findings"





Right AP view -Radiopaedia



Right Y view-Radiopaedia

Subacromial Bursitis, subacromial impingement, rotator cuff tendonitis/osis/tear

per UptoDate: In general, radiographs are unnecessary for the diagnosis but can be obtained to rule out other associated pathology.

Plain radiographs may be useful in the following clinical situations:

- No improvement with conservative therapy
- Evaluation of acromion morphology
- Evaluation of the acromioclavicular joint
- Evaluation of distance between acromion and humeral head
- Evaluation for tendon calcification
- Anatomical evaluation prior to subacromial or glenohumeral joint injection (not essential prior to injection)

Right AP view- with loss of subacromial space and osteoarthritic change-Radiopaedia



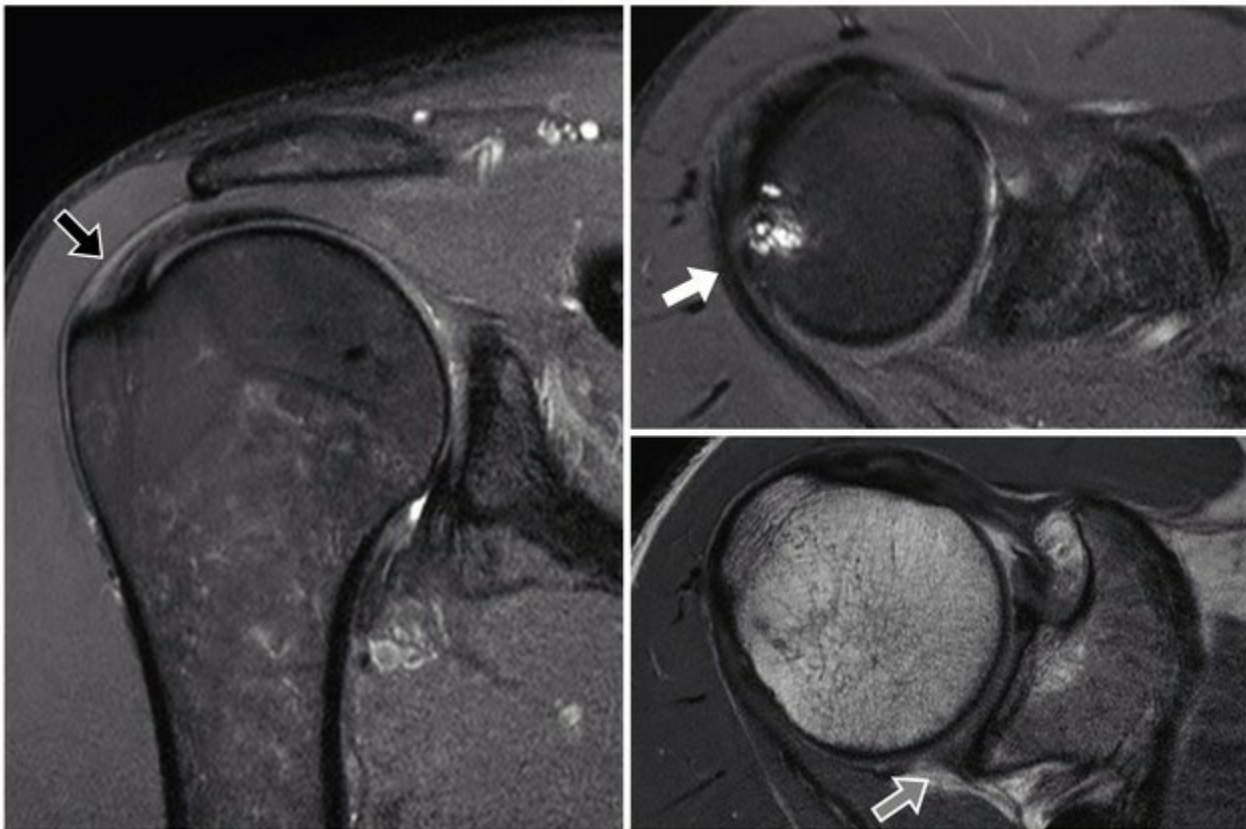
Magnetic resonance imaging — Magnetic resonance imaging (MRI) is most useful for assessing rotator cuff pathologies other than SIS, including labral tears, biceps tendon injuries, and intra-articular injuries that may benefit from surgery.

MRI is generally performed in the following circumstances:

- Symptoms and function fail to improve despite appropriate conservative therapy.
- Diagnosis remains unclear after initial evaluation.
- A rotator cuff or labrum tear is suspected based upon clinical presentation.

*Subacromial impingement is a dynamic condition not reliably diagnosed by MRI.

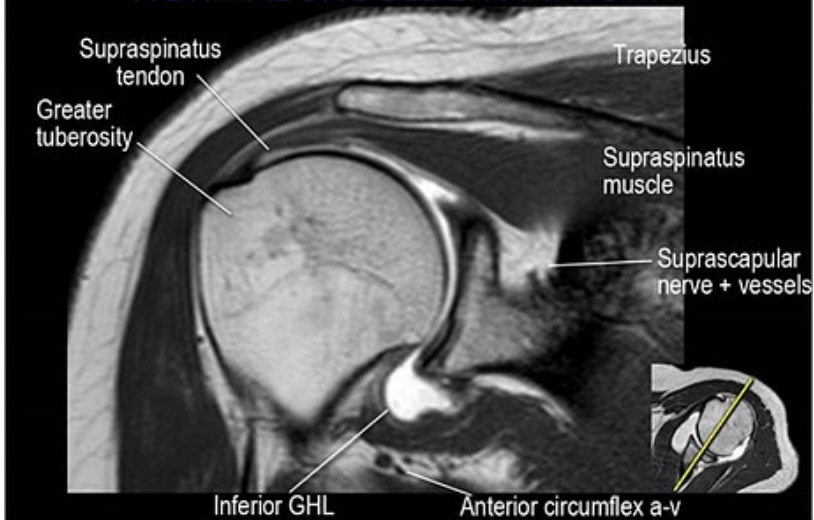
MRI images: Radiopaedia



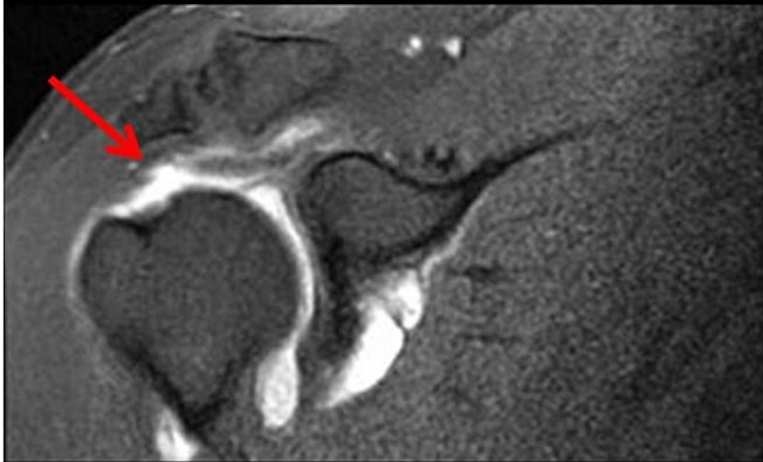
The acromioclavicular joint and glenohumeral joints are well aligned; there are moderate osteoarthritic changes involving the glenohumeral joint and the acromioclavicular joint. There is a loss of the subacromial space, suggesting subacromial impingement.

Images from Shields Health

NORMAL SHOULDER ANATOMY



SUPRSPINATUS TENDON TEAR



Images obtained from an MR arthrogram with contrast and indicate a tear in the rotator cuff.

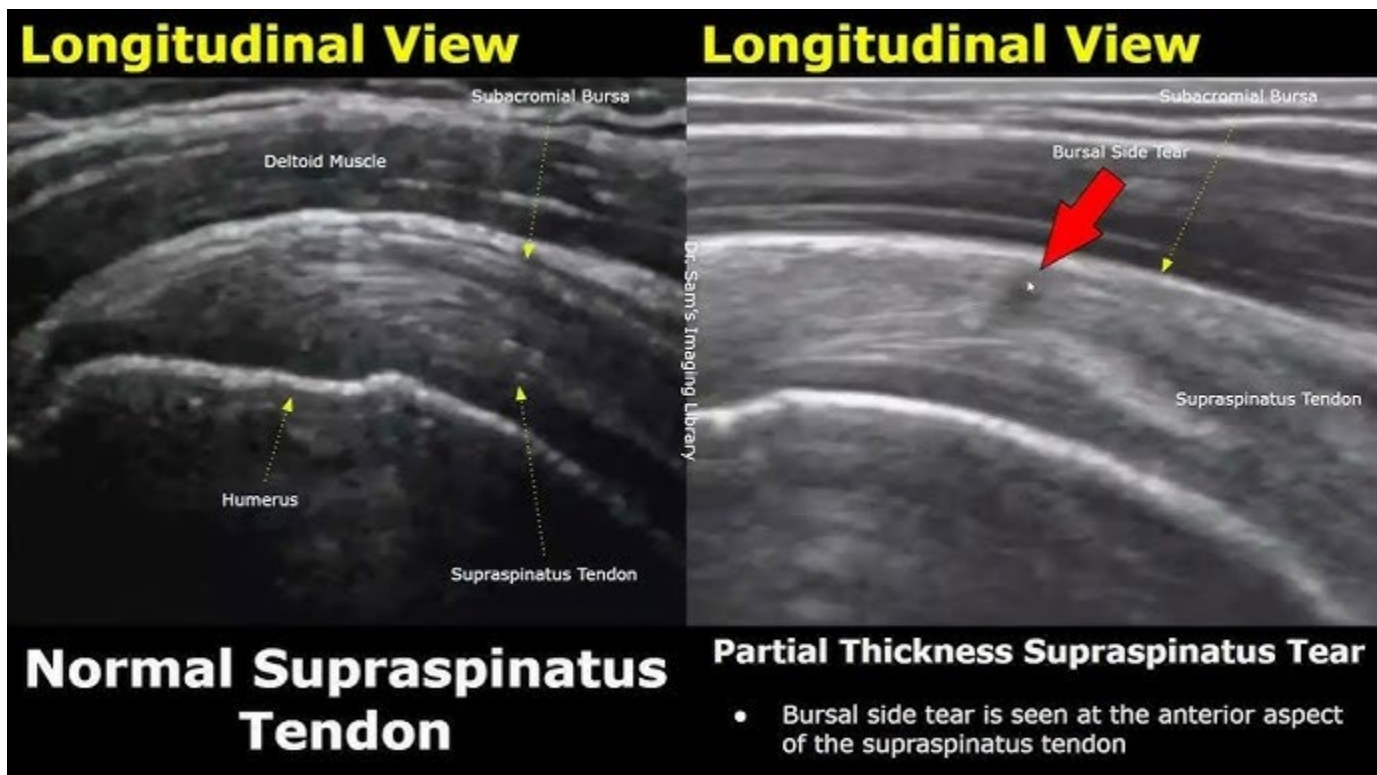
SUPRASPINATUS TENDINOSIS



Images obtained from a routine shoulder MRI , showing no tear to the rotator cuff, but does show tendinitis.

Can Also Consider US

Video-US Shoulder



Plan

Adhesive Capsulitis (Frozen Shoulder)-UptoDate

Often self-limiting though complete resolution may not occur

-Confirm diagnosis and rule out other causes of reduced motion/painful conditions at the glenohumeral joint

-X-ray of the shoulder to rule out other dx, US to rule out rotator cuff pathology

-Can consider subacromial injection w/lidocaine

-Can consider intraarticular cortisone injection for mod/severe symptoms

-Rest with gentle ROM

-Home exercises

-Pending response with conservative measures, consider PT, and/or referral to Orthopedics

Subacromial Impingement Syndrome and related issues-UptoDate

Rehabilitation of any injury requires a specific plan and exercise progression. With shoulder impingement syndrome (SIS), there are four primary goals of rehabilitation [[3,17-19Links to an external site.](#)]:

●Strengthen the muscles that stabilize the scapula: By strengthening the scapular stabilizers, greater stability is provided for the rotator cuff muscles, which originate on the scapula. This stability allows for greater efficiency and muscular endurance of the rotator cuff and improved overall shoulder function. This is a critical first step in rehabilitation. A subset of patients with deficiencies in scapular motion may need these addressed.

- Correct imbalances in strength and flexibility about the glenohumeral joint, including optimizing rotator cuff strength and ensuring mobility of the posterior shoulder capsule: Typically, before rehabilitation, the muscles at the front of the shoulder complex (anterior deltoid, internal rotator [ie, subscapularis]) are 1.5 to 2 times stronger than those at the posterior (posterior deltoid, external rotators).

- Stabilize the secondary movers of the shoulder complex: Once the primary muscles of the shoulder are strong and functional, the next step is to rehabilitate the secondary shoulder muscles in order to improve coordination of the entire shoulder complex.

- Correct imbalances, weaknesses, or dysfunction at any segment along the kinetic chain (ankle, knee, hip, torso, etc).

In addition to these four primary goals, a fifth goal for many athletes is to improve sport-specific biomechanics and function. This may entail performing exercises that simulate key movements in their sport using resistance or other techniques to improve performance. Proper technique is an essential component of this phase. During this final stage, it may be helpful to have a coach with sport-specific expertise assess the athlete's mechanics.

Patients should be made aware that successful completion of such a program generally requires 8 to 16 weeks, but some improvement is usually noted within the first three to four weeks. A patient who has successfully completed a rehabilitation program for SIS should have complete, pain-free motion of the glenohumeral joint and should be able to perform all functional movements and exercises in the program without pain. Several months following the completion of rehabilitation may be needed before pain during sleep completely resolves. For athletes, successful rehabilitation is generally determined by the ability to return to play at their previous level of performance.

It is important to maintain the patient's confidence during rehabilitation. If a program is too easy and provides little benefit, patient compliance may fall; if a program is too difficult, pain may increase and the patient may quit rehabilitation. To avoid such problems when rehabilitating patients with shoulder pain, it is important to distinguish between SIS and rotator cuff tears at the outset and to refer patients back to their primary care or sports medicine clinician to discuss possible surgical evaluation as soon as the need is recognized (eg, patient demonstrates significant weakness consistent with a substantial rotator cuff tear).

SLAP tears-UptoDate

Initial treatment — Initial management of SLAP tears involves reducing pain by avoiding aggravating activities, such as overhead throwing or work where the shoulder is placed repeatedly in abducted and externally rotated positions. Nonsteroidal antiinflammatory drugs (NSAIDs) or [acetaminophen](#) can be taken to help reduce pain in the acute setting. Once pain is adequately controlled, patients begin a rehabilitation program with the goals of improving glenohumeral and scapulothoracic motion and increasing the strength and endurance of the rotator cuff and scapulothoracic muscles [[28Links to an external site.](#)].

Nonoperative treatment — Typically, nonoperative management of SLAP tears is preferred whenever possible given the long recovery required following surgical repair (typically 6 to 12 months) and the

limitations of surgical treatment [[45Links to an external site.](#)]. Rehabilitation programs focused specifically on addressing scapular dyskinesia and posterior capsule contractures associated with glenohumeral internal rotation deficit (GIRD) enable approximately 40 percent of professional baseball players to resume playing without surgery [[46Links to an external site.](#)]. Although studies are limited primarily to retrospective case series, evidence suggests that most patients are able to resume preinjury activity levels following participation in a well-designed physical therapy program.

In a systematic review of five studies involving 244 young adult athletes with primarily type 2 SLAP tears, the return-to-play rate was approximately 50 percent overall and 76 to 78 percent among those who completed the rehabilitation protocol, which required 20 sessions on average [[47Links to an external site.](#)]. As an example of the type of study included in the review, a retrospective study of 39 patients involving a rehabilitation protocol similar to that described above for baseball players reported improved pain and quality of life scores in 71 percent of patients, and 67 percent of overhead athletes returned to their sport at preinjury levels [[48Links to an external site.](#)].

The typical approach to rehabilitation of SLAP lesions includes addressing the following principles and biomechanical deficits [[8,49,50Links to an external site.](#)]:

- Reduce pain and inflammation.
- Restore pain-free range of motion, including identifying and resolving any GIRD. Specific stretches and exercises help improve internal rotation in these patients.
- Improve function and strength of scapular stabilizers – Exercises to accomplish this goal are described separately.
- Improve rotator cuff strength – Exercises to accomplish this goal are described separately.

Osteopathic Considerations discussed next!!

Osteopathic Considerations (ABC'S)

- **A (Autonomics)**
 - **Sympathetics**
 - Originate at T1-T4- innervation then goes to the cervical ganglia before making its way to the upper extremities
 - **Parasympathetics**
 - Few in the extremities
 - **Other Innervation**
 - Brachial plexus- Consider the innervation of the muscles of the shoulder and where those nerves may become impinged or entrapped- be sure to evaluate the cervical spine

- **B (Biomechanics)**

- Consider that restrictions in one joint can lead to compensatory movement in other joints. All joints of the shoulder need to move freely in order for proper functioning of the shoulder.
- “Some factors that predispose the patient to shoulder pain include unstable glenohumeral joints, weakness of scapular stabilizers, abnormal posture, and hypomobility of the cervical and thoracic spine” (Pain Medicine & Management 2nd edition)
- Consider the importance of motion in each of the following joints:
 - ***Sternoclavicular joint***
 - Important connection of the upper extremity to the thorax- the only true bony joint to do so. Restriction of motion here will lead to restriction of motion at the AC joint.
 - Shoulder shrug = inferior glide
 - Shoulder depression = superior glide
 - Scapular protraction = posterior glide
 - Scapular retraction = anterior glide
 - ***Acromioclavicular joint***
 - Connects the scapula to the clavicle and motion is necessary for shoulder abduction to occur.
 - Adduction causes superior glide
 - Abduction causes inferior glide
 - Internal rotation causes anterior glide
 - External rotation causes posterior glide
 - ***Glenohumeral joint***
 - Lots of mobility, little stability. The rotator cuff muscles work to keep it in place and stabilized. Alteration in rotator cuff function or scapular kinematics can affect this joint.
 - Commonly the posterior capsule is tight- this causes anterior translation and elevation of the humeral head which could lead to impingement syndromes.
 - Major motions: Flexion, Extension, abduction, adduction, IR, EX
 - Minor motions: superior/inferior and anterior posterior glides
 - ***Scapulothoracic joint***
 - The scapula on the thorax- pseudoarticulation- attached by rhomboids, trapezius, serratus anterior, pectoralis minor and levator scapula. Tension on the muscles controls the position. Changes in the position of the scapula alter the length tension relationships and positions of the rotator cuff muscles and joint arthrokinematics.
 - Motions include superior/inferior glide, rotation and protraction/retraction

- **Elbow**
 - Connections from the shoulder (biceps, triceps) have long connections to the elbow. Elbow biomechanics may play a role in shoulder biomechanics.
 - Consider the:
 - Radiohumeral joint (anterior/posterior glide with supination and pronation)
 - Ulnohumeral joint (abduction/adduction glide and slide)
- **Thoracic spine/Ribs**
 - The scapula must glide over the thoracic spine and ribs. Somatic dysfunction of the thoracic spine may alter the position of the scapula or affect musculature connecting to the scapula.
- **Cervical spine**
 - Many of the muscles that control scapular motion attach or originate from the cervical spine, therefore these areas must be assessed for somatic dysfunction. Additionally the brachial plexus comes from C5-T1 and should be assessed for anything that may impede or compress it.
- **Pelvis**
 - Don't miss long musculoskeletal and fascial connections like the latissimus dorsi which attaches onto the iliac crest and the humerus
- **C (Circulation)**
 - **Axillary region**- lymphatics from the upper extremity flow to the axillary region before moving into the thoracic duct. Myofascial restrictions in this region, including from pectoralis major/minor may impede lymphatic flow
 - **Thoracic inlet**- lymphatic and vascular drainage from the arm may be impeded here
 - **Diaphragm**- major lymph/venous pump
- **S (Screening)**
 - **Postural issue**- forward head/neck posture, anterior shoulder, scoliosis
 - **Upper Crossed Syndrome**
 - Typically tight: pectoralis major, upper trapezius, levator scapulae, scalenes, sternocleidomastoid, latissimus dorsi, biceps, supscapularis
 - Typically weak: triceps, rhomboids, serratus anterior, lower trapezius, supraspinatus, infraspinatus
 - **Leg length discrepancy- or sacral base unleveling can cause compensatory postures and altered biomechanics all the way up to the thorax, ribs, cervical spine head and even upper extremity**
 - Mind/Body/Spirit Considerations- life stressors can contribute to increase in muscle tensions in the neck and shoulder region
 - **Ergonomics**- how does this patient sit, stand, work, sleep

Osteopathic Treatment

Osteopathic Diagnosis and Treatment Considerations

Handout:

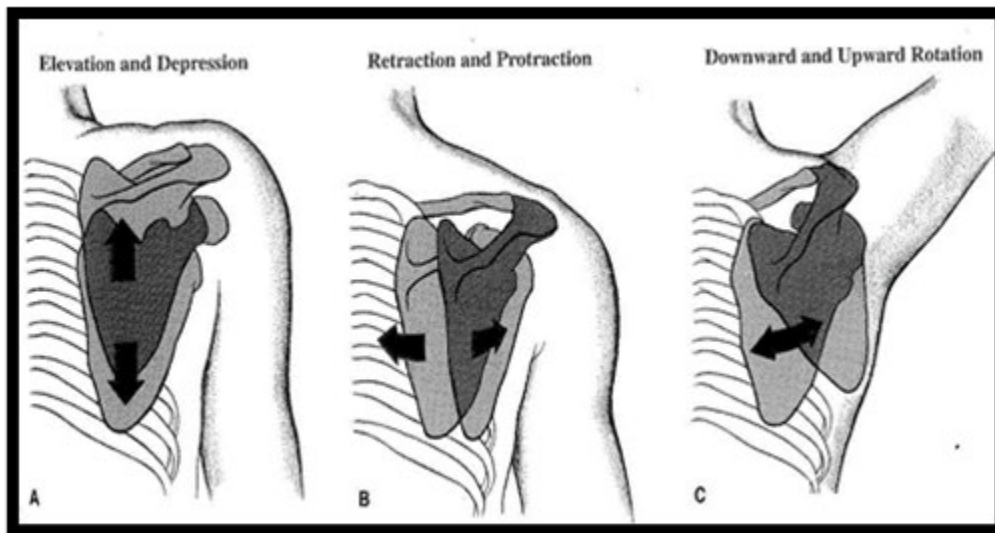
ABC's

When considering the upper extremity be sure that you consider the ABC's.

- **Autonomics**
 - T1-T5
- **Biomechanics**
 - Gross range of motion
 - Glides and slides of each joint
 - Upper crossed syndrome considerations
- **Circulation**
 - thoracic inlet, diaphragm

DIAGNOSIS:

Scapulothoracic Joint:



- Observe for static positional asymmetry/ deformity(winging)
- Palpate
- Ortho component-Active Range of Motion (AROM) testing for instability/gross motion restriction
- 3 Motions:
 - Rotation about an A-P axis – Use to lateralize to most restricted side
 - Created by Shoulder Abduction:(“Normal ROM”=180 Deg.) palms facing out.
 - Observe & palpate for asymmetry & quality of motion. The side that moves first is considered to be the most restricted- as described in prior labs. Observe all the way to the fingers for static asymmetry
 - Superior (aka upward) rotation=glenoid fossa facing superior-Abduct arm
 - Inferior (aka downward) rotation=glenoid fossa facing inferior- Adduct arm
 - Elevation/Depression (Shrug shoulders, reach towards floor)
 - Protraction/Retraction (move shoulders anteriorly & posteriorly toward midline of body)

Glenohumeral Joint:



Passive ROM- minor motions-Short-lever

- Stabilize Glenoid fossa by contacting scapular spine & clavicle/acromion
- Grasp proximal humerus
- Ortho component- Motion test humeral head gently but firmly to end ROM of joint capsule, with respect to Glenoid fossa
 - Superior/Inferior Glide
 - feel for instability (Sulcus sign) during **gross** Inferior Glide
 - Anterior/Posterior Glide
 - feel for instability during **gross** A-P Glides
- Is gross gliding the same as the gliding used to assess somatic dysfunction?

AC Joint:

- Patient seated- operator behind the patient- one hand palpates the superior aspect of the AC- the other grasps the forearm
- Operator, while monitoring at the AC Joint:
 - Adduction causes superior glide
 - Abduction causes inferior glide
 - Internal rotation causes anterior glide
 - External rotation causes posterior glide
- Restriction in symmetric motion denotes somatic dysfunction

SC Joint:

1. Observe for static positional asymmetry/deformity
2. Palpate SC joints & note Asymmetry of position, Tissue Texture, and Tenderness
3. Spring SC Joints to help lateralize: GENTLY spring proximal clavicles inferiorly, superiorly and posteriorly (segmentally diagnose and treat side with less spring first)
 - a. Inferior/Superior glide (have patient shrug shoulders, then reach towards their feet)
 - b. Posterior/Anterior glide (shoulder flexion to 90 deg. then reach forward) Protraction about 2-3 inches, followed by Retraction)
 - c. Anterior/Posterior rotation (abduct arms to 90 deg., Internally Rotate humerus until clavicle anteriorly rotates, Externally Rotate humerus until clavicle posteriorly rotates -with elbows flexed or straight)

Abducted/Adducted Elbow:

1. Patient is supine and shoulder abducted to 45 degrees. Place the hand midway between supination and pronation (you can then pin it at your side). Grasp the elbow with both hands, induce slight flexion
2. Introduce a very slight medial and lateral motion and observe for ease and tension paying close attention to anatomic barriers.

Radial head:

1. Use Handshake grip
2. Other hand fingers or thumb monitors radial head for response to:
 - a. Supination-radial head glides anteriorly
 - b. Pronation- radial head glides posteriorly

Interosseous membrane forearm

1. Palpate Interosseous membrane for region of greatest Tissue tension/tenderness
2. Start proximally on posterior (dorsal) aspect of forearm just distal to Radial head and move distally. once tissue thins out you can palpate either anterior or posterior aspects of distal forearm and reach the IOM

TREATMENT:

Autonomics and Circulation

Soft tissue to the Thorax/Lumbar

Rib Raising - seated or supine

Thoracic HVLA - Texas Twist (flexed only)

Diaphragm MFR

Thoracic inlet MFR

Biomechanics

Scapulothoracic MFR/ME (Scapular lift)

1. Patient is lateral recumbent, operator faces the patient. Internally rotate their arm and place it behind their back.
2. Operators cephalad hand stabilizes the shoulder and caudad hand grasps the medial border of the scapula
3. Lifting the scapula away from the rib cage, you can treat myofascially or engage restrictive barriers with muscle energy. Respiratory cooperation can also be used.

Seven Stages of Spencer

The patient is side-lying- utilize muscle energy principles- move into the motion barrier- have the patient isometrically contract in the other direction, relaxation, engage the next barrier

1. Extension
2. Flexion
3. Circumduction with compression
4. Circumduction with traction
5. Abduction
6. Internal rotation
7. Joint pump

Can do Seated, or Lying Lateral Recumbent



ME for the AC Joint

1. Patient is seated- operator stands in the same position as when diagnosing (as listed above)
2. Position the patient into the restrictive barriers (restricted glides)
3. Patient utilizes ME pushing in the direction opposite of the restrictive barrier
4. Operator takes up the slack by advancing further into the direction of the restrictive barrier

Glenohumeral BLT/MFR

Patient seated or supine, if performed seated patient may support elbow with their opposite hand.

1. Use same grips as for segmental diagnosis
2. Operator places the humeral head into its motions of ease to balanced tension
3. Can also be performed as a long lever technique by moving distal humerus as described above for Dx

1st rib FPR

- Pt is supine- operator stands facing them- placing monitoring hand (the one nearest patient) on the posterior first rib to monitor
- With the opposite hand, the operator grasps the elbow, and abducts the arm to 90 degrees with full elbow flexion
- A compressive force is created through the elbow downward to the monitoring finger
- Operator uses their monitoring arm as a fulcrum, increasing shoulder flexion- holds for 3-5 seconds
- Maintaining the compressive force and IR, the operator adducts the upper arm across the patient's chest and swing it down into a neutral position

Counterstrain Tenderpoints



Long head of biceps

F ABd
IR

ELBOW: flex

SHOULDER: flex + minimal
ABd, IR

Biceps tendon in the
bicipital groove



Short head of biceps/coracobrachialis IR

ELBOW: flex

SHOULDER: flex + ADd + IR

At the inferolateral aspect of the coracoid process



Supraspinatus

F ABd
ER

SHOULDER: flex + ABd + marked ER

In the belly of the supraspinatus muscle

Elbow

- **Elbow Wobble (Articulatory)**

- Patient is supine and shoulder abducted to 45 degrees. Operator places the hand midway between supination and pronation (then can pin it at the operator's side). Grasp the elbow with both hands
- Operator produces a wobble in a medial/lateral direction while flexing and extending in order to produce articulation at the elbow

- **Abducted/adducted elbow**

- Same positioning as above

Clavicle treatment- BLT

- The patient is seated- operator stands in a split stance in front of the patient with one hand grasping the distal clavicle and one hand grasping the proximal clavicle
- Operator then has the patient lean forward until the operator feels that they can "float" the clavicle
- The operator can modify their pressure to lift either end up or down, IR or ER, compression or distraction until a balance point of even tension is reached
- Maintain this hold until a release is felt. Can fine tune by having the patient very slowly turning their head to the left or to the right
- After the release- the operator supports the clavicle as the patient sits back up- releasing only after they are back in a normal seated posture

Pharmacology

Acute treatment — Research to direct management of Shoulder Impingement Syndromes is limited, but the following is a generally accepted approach for acute symptoms: per UptoDate

- Cryotherapy – Ice may decrease acute swelling and inflammation and provide some analgesia.
- Rest – This means avoiding activities that aggravate symptoms, including all overhead activities.
- Nonsteroidal anti-inflammatory drugs (NSAIDs)

For acute injuries, we give a short course (two to four weeks) of scheduled NSAID therapy. If the patient has no major contraindications to high-dose NSAIDs (eg, bleeding gastric ulcer), we often prescribe a two-week regimen (eg, [ibuprofen](#)[Links to an external site.](#) 800 mg three times daily, [naproxen](#)[Links to an external site.](#) 500 mg twice daily). If there are relative contraindications to high-dose NSAIDs (eg, mild renal insufficiency), we may give a two-week course of an over-the-counter NSAID using a lower dose (eg, ibuprofen 200 to 400 mg). Thereafter, patients may use an NSAID as needed.

Table 1.
Common Oral NSAIDs

Generic	Brand	Max Daily Dose
OTC		
Ibuprofen	Advil, Motrin	1,200 mg
Naproxen	Aleve	1,500 mg
Prescription		
Celecoxib	Celebrex	400 mg
Diclofenac	Voltaren	200 mg
Diffunisal	N/A	1,500 mg
Etodolac	N/A	1,000 mg
Flurbiprofen	Ansaid	300 mg
Ibuprofen	N/A	3,200 mg
Indomethacin	Indocin	200 mg
Ketoprofen	N/A	300 mg
Ketorolac	Toradol	40 mg (max 5 days of total treatment)
Meloxicam	Mobic	15 mg
Nabumetone	N/A	2,000 mg
Naproxen	Naprosyn	1,500 mg
Oxaprozin	Daypro	1,800 mg
Piroxicam	Feldene	20 mg
Sulindac	Clinoril	400 mg

max: maximum; N/A: not applicable; NSAID: nonsteroidal anti-inflammatory drug. Source: References 7, 22.

Table 2.
Risk of GI Toxicity

High Risk
History of a previously complicated ulcer, especially recently
Multiple (>2) risk factors
Moderate Risk (1-2 risk factors)
Age >65 y
High-dose NSAID therapy
A previous history of an uncomplicated ulcer
Concurrent use of aspirin (including low-dose), corticosteroids, or anticoagulants
Low Risk
No risk factors

GI: gastrointestinal; NSAID: nonsteroidal anti-inflammatory drug. Source: Reference 15.

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